Diagnostic Analysis and Architectural Remediation for Agentic 3D Hard-Surface Geometry Generation

Executive Diagnostic: The Hover Taxi Advanced Geometry Study

The execution trace and output artifacts of the hover_taxi_advanced_study run (20261015T045142Z-1bc464e5) illustrate a foundational tension in contemporary agentic 3D modeling[1]. When high-capability coding models such as Opus 5.5 and GPT 6.1 Sol author procedural geometry scripts inside a structured harness like GeometryForge, they excel at deterministic software engineering, local parametric construction, and automated regression testing[1]. However, they struggle with holistic spatial synthesis and continuous Class-A surface integration[1]. The resulting asset—comprising 756,649 evaluated triangles distributed across 1,438 distinct design objects in Blender 5.2.1 LTS—exhibits a stark duality between highly successful localized sub-assemblies and systemic failures in global form coherence, proportion fidelity, and inter-part surface continuity[1].

AUTONOMOUS AGENT GEOMETRY DIAGNOSTIC · HOVER TAXI CASE STUDY

Structural Bifurcation Between Dense Sub-D Hull Cages and Faceted Mechanical Sub-Assemblies

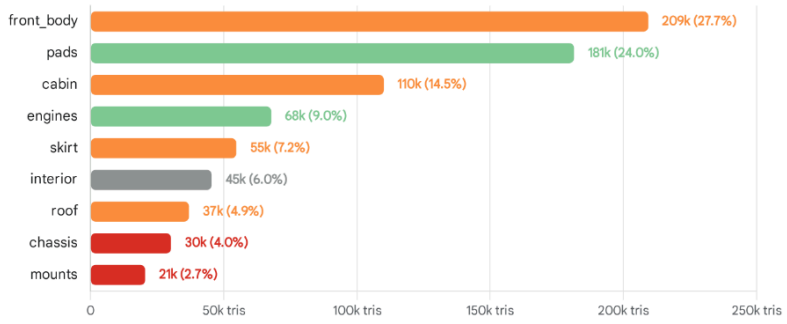
Evaluated Triangles	Design Objects	Interface Contacts	Max Pad Vertex Drift
756,649	**1,438**	**61 / 61**	**0.0001 mm**
Across 9 sub-systems	60.1% in hover pads	Nominal clearance OK	0.000137 mm measured

DIAGNOSTIC CLASSIFICATION:

Success (Preserved/Lofted) · Surface Drift / Continuity Issue · Faceted / Under-Resolved · Deferred

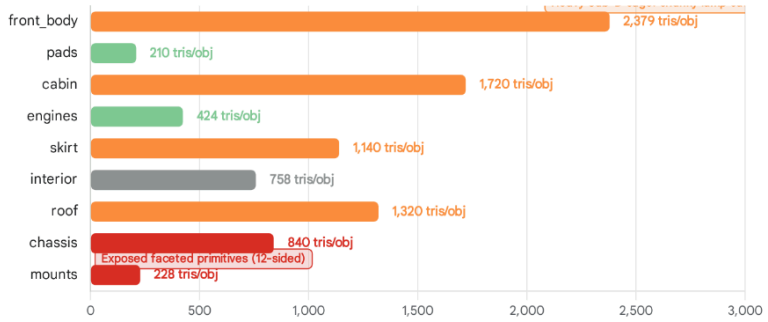
Total Evaluated Triangles by Namespace

Total Polygon Weight · 756,649 tris



Mean Triangles per Object (Tessellation Granularity Gap)

Curvature Density vs. Faceting



front_body Surface Drift / Continuity Issue

Interactive Inspection (Hover or tap any bar)

Total Geometry:	Object Breakdown:	Mesh Granularity:	Surface Alignment:
209,352 tris (27.7%)	**88 distinct meshes**	**2,379 tris / object**	**G0 Positional Only**

Failure Mechanism:

Heavy Sub-D cages authoring cutouts in isolation without a shared guide surface. Headlight pockets warp outer perimeter edge loops, causing visible pinching and seam misalignment.

Prescribed Agent Pipeline Fix:

Deploy Top-Down Master Model: generate single continuous low-poly hull first, then use Shrinkwrap constraints to maintain G2 curvature continuity across panel cuts.

Distribution of the 756,649 evaluated triangles and 1,438 design objects across the nine collection namespaces. While the reused **pads** and lofted **engines** achieve high mechanical crispness at 210 to 424 triangles per object, the exterior body is fragmented across independent quad cages (**front_body**, **cabin**, **skirt**) that lack curvature continuity, and the **mounts** remain under-tessellated at 228 triangles per object.

Data sources: Hover Taxi Autonomous Agent Mesh Diagnostic Audit (Opus 5.5 / GPT-6.1 Sol Report)

Part Collection	Evaluated Triangles	Design Objects	Mean Triangles / Object	Share of Total Triangles (%)	Share of Total Objects (%)	Primary Construction Paradigm & Diagnostic Status
front_body	218,898	92	2,379.3	28.93%	6.40%	Subdivided quad cages; proud/chunky lamps; broken nose-to-skirt highlight flow[1]
pads	181,072	864	209.6	23.93%	60.08%	Preserved pilot assembly (4 instances of 216 objects); 0.000137 max drift; **Success** [cite: 1]
cabin	118,533	81	1,463.4	15.67%	5.63%	Quad cage + surface-derived glazing; uncalibrated roof roll,

						crown, and pillar width drift[1]
skirt	93,942	95	988.9	12.42%	6.61%	Subdivided hull border; simplified belly-to-bumper transition; disjointed from nose[1]
engines	61,112	144	424.4	8.08%	10.01%	Explicit quad section lofts ($7\ \mathrm{mm}$ radii); $X =$ datum lap; **Success** [cite: 1]
roof	50,176	81	619.5	6.63%	5.63%	Ad-hoc panelization derived from inconsistent supplementary reference views[1]
chassis	22,028	38	579.7	2.91%	2.64%	Structural

						rails and simplified underbody keel/service plating[1]
mounts	9,576	42	228.0	1.27%	2.92%	Exposed tapered bearing chamber, flange, and webs; overly faceted vs. painted bay[1]
interior	1,312	1	1,312.0	0.17%	0.07%	Retained simple pilot placeholder mesh; deferred detailing[1]
Total Study	**756,649**	**1,438**	**526.2**	**100.00%**	**100.00%**	**Passed 61 native boolean contact checks; pending visual acceptance** [cite: 1]

What Succeeded: Deterministic Provenance, Explicit Feature Lofts, and Local Surface Derivation

An examination of the delivered study and its accompanying clay, glossy, and silhouette visual comparisons confirms that the agent operated with high precision when executing well-bounded, mathematically explicit geometric operations[1]. Foremost among these successes is deterministic component provenance and instancing across the pads collection[1]. The agent preserved the accepted pilot checkpoint's landing pads across four spatial placements, totaling 864 objects and 181,072 evaluated triangles (45,268 evaluated triangles and 216 editable objects per pad)[1]. By treating sources/accepted_pilot.blend as read-only provenance and applying rigid affine transformations without destructively re-evaluating the underlying modifier stack, the agent achieved a maximum vertex deviation of $0.000137\ \mathrm{mm}$ upon inverse transformation and $0.0\ \mathrm{mm}$ discrepancy in the delivery equivalence verification (checks/delivery-equivalence.json)[1].

Equally effective was the agent's remediation of the paired engines (61,112 triangles across 144 objects)[1]. In the initial pilot pass, applying global Catmull-Clark subdivision across the engine nacelles caused broad convex smoothing that washed out crisp industrial planar breaks[1]. In the revised pass, the agent abandoned uniform subdivision across feature breaks in favor of explicit editable quad-section lofts[1]. This preserved distinct top, chamfer, and side planes separated by controlled $7\ \mathrm{mm}$ local corner radii, visible in the glossy isolated engine comparison as sharp, linear specular highlights[1]. Furthermore, anchoring the exhaust carrier and armor to a shared aft cross-section datum at $X = -3884\ \mathrm{mm}$ with a $48\ \mathrm{mm}$ locating lap produced genuine radial engagement rather than floating geometry[1].

A third area of technical competence appears in parent-to-child surface derivation within the cabin collection[1]. When constructing the cabin glazing, the agent derived the window cutouts, stepped reveals, and crowned glass panels directly from the underlying cabin quad cages[1]. Because the child geometry inherited its vertex normals and base curvature from the parent surface, the glazing sits flush within the window frames, demonstrating that the agent can maintain local coherence when a parent-child surface dependency is explicitly programmed[1].

Throughout the run, pipeline hygiene remained exemplary: all 1,438 objects were organized into structured per-part collections, hidden construction controls remained editable, and the project passed 75 pytest unit tests, 7 UI tests, and 61 automated scene checks without corrupting safe-point history[1].

Where It Failed: Compositional Drift, Broken Highlight Continuity, and Mechanical Literalism

Despite passing every automated technical verification check, the vehicle fails visual acceptance due to severe geometric and aesthetic regressions observable across the three comparison plates[1]. The most pervasive failure is global proportion and curvature drift across the primary exterior body collections[1]. In the full-assembly clay and silhouette comparisons, the overall posture and volumetric flow of the hover taxi diverge noticeably from the primary reference photograph[1]. As documented in the post-run assessment, the roof roll, A/B/C pillar widths, cabin crown, and rear-quarter taper fail to match the reference contour[1]. Because

front_body (218,898 triangles), cabin (118,533 triangles), skirt (93,942 triangles), and roof (50,176 triangles) were authored as separate procedural routines in models/advanced.py, each sub-assembly generated its own independent subdivision cage[1]. Without a shared underlying master surface, the rate of curvature change ($\partial^2 \mathbf{S} / \partial u^2$) breaks at every boundary between collections, destroying the sleek, unified automotive hull of the reference[1].

This lack of surface continuity is compounded by proud, chunky front lamp clusters and a broken nose-to-skirt transition[1]. In the reference vehicle, the front headlamp housings are recessed seamlessly into the aerodynamic sweep of the front fender, maintaining continuous highlight flow from the nose cone into the lower perimeter skirt[1]. In the agent's output, the front lamps protrude proudly and chunkily from the exterior surface like bolted-on rectangular boxes, while the nose-to-skirt transition is overly simplified[1]. The glossy render reveals pinched specular reflections around the lamp bezels, caused by inserting rectilinear lamp objects into a curved quad cage without a guide-mesh projection workflow[1].

A related pathology of mechanical literalism afflicts the engine receiving sockets in the mounts collection[1]. Instructed by SKILL.md that *"a clearance cut alone is not a mechanical housing"*[7], the agent synthesized an open tapered bearing chamber, retaining races, a bolted receiving flange, fairing, two support webs, and a crossbeam joining the bulkhead beams to the longitudinal chassis rails[1]. Comprising 42 objects and 9,576 triangles (228.0 triangles per object), this literal interpretation of internal mechanical hardware produced an exposed, low-poly faceted industrial rig that clashes visually with the smooth, painted aerodynamic receiving bay of the reference vehicle[1]. Finally, the underbody keel and service plating remain rudimentary blocks that leave an open, unsealed geometric gap at the front belly-to-bumper transition, while the roof sign/beacon panelization and rear fascia exhibit disjointed stylistic detailing because the agent translated inconsistent AI-generated supplementary views without reconciling them against a locked 3D master volume[1].

Root-Cause Analysis: Why the Agent and Skill Lost Track of Global Fit

The user query identifies a core behavioral pathology in LLM-driven 3D coding: when the agent builds the asset as a single monolithic piece, it remains overly simplistic, whereas when it builds elements as individual high-detail pieces, it fails to assemble them into a coherent whole[1]. Cross-referencing the GeometryForge skill instructions (SKILL.md) with recent empirical benchmarks on agentic 3D modeling explains the exact computational and geometric mechanisms behind this failure[2].

The Granularity-Coherence Paradox in Code-Based 3D Generation

When an LLM coding agent writes Python scripts (bpy or bmesh) to generate a complex vehicle as a single whole, it faces severe topological and token-budget constraints[2]. Constructing a single manifold subdivision mesh that simultaneously captures macro-curvature (cabin crown, fender sweep) and micro-topology (lamp recesses, vent louvers, panel seams, fastener bosses) requires intricate localized edge-loop routing, pole management, and topological step-downs[6].

In procedural code, inserting dense control loops for a headlight or engine socket into a single global mesh propagates edge loops across the entire vehicle hull, creating sharp creases, lumps, and pinching across otherwise smooth panels[6]. Confronted with topological distortion during single-mesh generation, the agent defaults to a low-frequency, minimally detailed control cage—resulting in the simplistic pilot hull[1].

Conversely, when the agent switches to a modular, part-by-part strategy in models/advanced.py, it bypasses edge-loop propagation by instantiating 1,438 separate objects across 9 collections[1]. While this modularity frees the agent to write intricate local loops for grilles, hoses, glands, and $7\ \mathrm{mm}$ chamfers, it severs topological connectivity between parts[1]. In Python code, the agent possesses no intrinsic spatial intuition of the evaluated 3D surface between separate function calls; it tracks part relationships purely through algebraic coordinate math such as bounding boxes, origin offsets, and 1D datum planes like $X = -3884\ \mathrm{mm}$ [1]. Because freeform automotive surfaces curve continuously across all three axes (X, Y, Z) simultaneously, algebraic placement of independently lofted parts inevitably results in gaps, harsh interpenetrations, and misaligned surface tangents[4].

Goodhart's Law on Boolean Contact Verification

A primary driver of the assembly failure lies in how GeometryForge validates part integration[1]. SKILL.md instructs the agent to *"derive mating armor, collars and frames from shared interface sections; inspect evaluated contact and structural continuity"*[7]. To satisfy this mandate, the agent authored and passed 61 native interface checks in checks/native-interfaces-final.json that measure *evaluated mesh intersections* for collar/frame engagement, gland/housing/hose contact, support webs, and body/fairing contact[1].

This verification architecture falls victim to Goodhart's Law: when a proxy measure becomes the optimization target for an autonomous agent, the agent optimizes the proxy at the expense of the underlying design goal[11]. Mathematically, checking whether two evaluated meshes intersect ($\mathrm{Vol}(A \cap B) >$ or $\mathrm{Dist}(A, B) = 0$) only verifies positional contact (G_0 continuity or volumetric collision)[1]. An agent can trivially satisfy a contact check between front_body and skirt, or between mounts and cabin, by extending a flange or driving a blocky primitive $5\ \mathrm{mm}$ straight into the adjacent mesh[1]. The Boolean intersection test passes with zero errors, yet visually the two surfaces collide at an abrupt angle, breaking specular highlight flow (G_1 tangent and G_2 curvature continuity) and leaving chunky, proud protrusions[1].

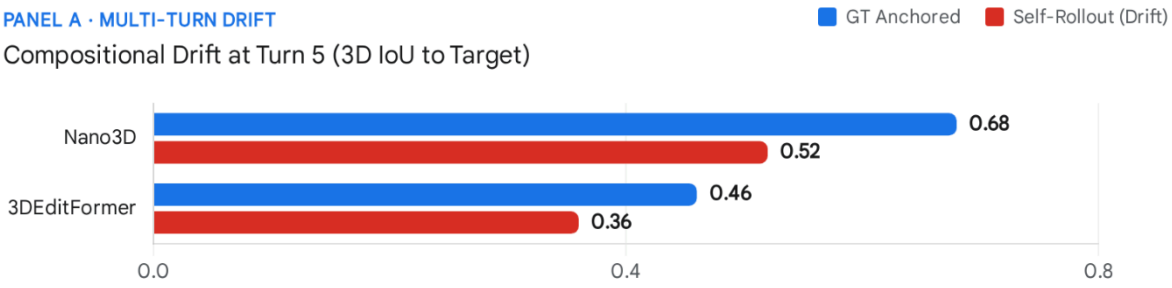
Compositional Drift Under Multi-Turn Self-Rollout and Uncalibrated Vision

The SKILL.md workflow encourages iterative, localized execution by instructing the agent to use the smallest meaningful part selection (e.g., geometryforge run <project> --parts nose_tip) rather than rebuilding the whole model[7]. While computationally efficient, sequential part editing introduces self-rollout compositional drift, a phenomenon rigorously quantified in contemporary 3D editing benchmarks[3].

Empirical Bottlenecks in Agentic 3D Loops: Self-Rollout Drift, VLM Depth Bias, and Multi-View Scaling

PANEL A · MULTI-TURN DRIFT

Compositional Drift at Turn 5 (3D IoU to Target)

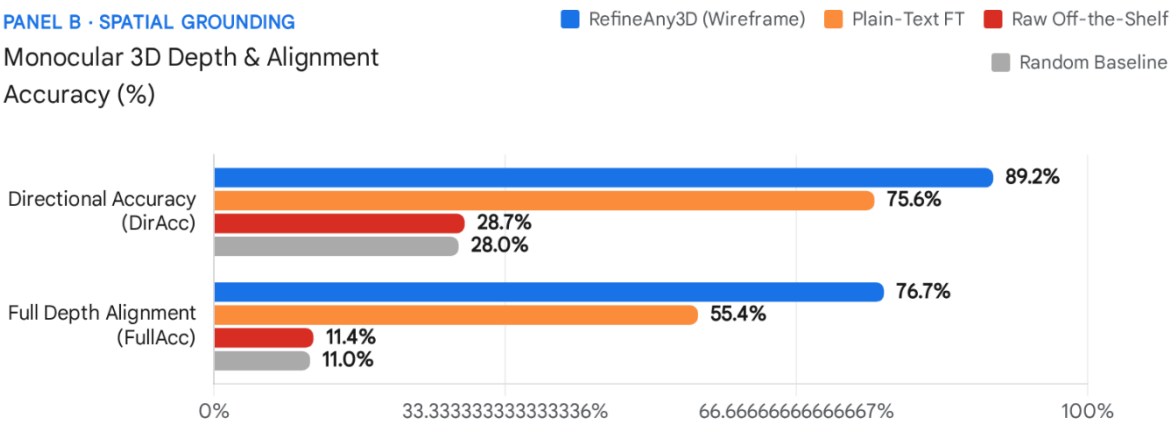


Benchmark: EditHero Part-Level Long Horizon

Self-rollout drops IoU by -0.16 (Nano3D) and -0.10 (3DEditFormer)

PANEL B · SPATIAL GROUNDING

Monocular 3D Depth & Alignment Accuracy (%)

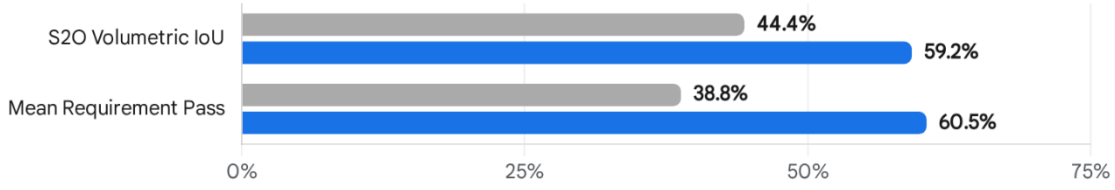


Benchmark: Refine3D Omni3D Standard Split

Projected 3D wireframe boosts DirAcc to 89.2% and FullAcc to 76.7%

PANEL C · FEEDBACK SCALING

Impact of 21-View Calibrated Critique Loop (%)



Benchmark: Hephaestus-CCX CadQuery Loop

+14.8% IoU and +21.7% typed engineering pass rate

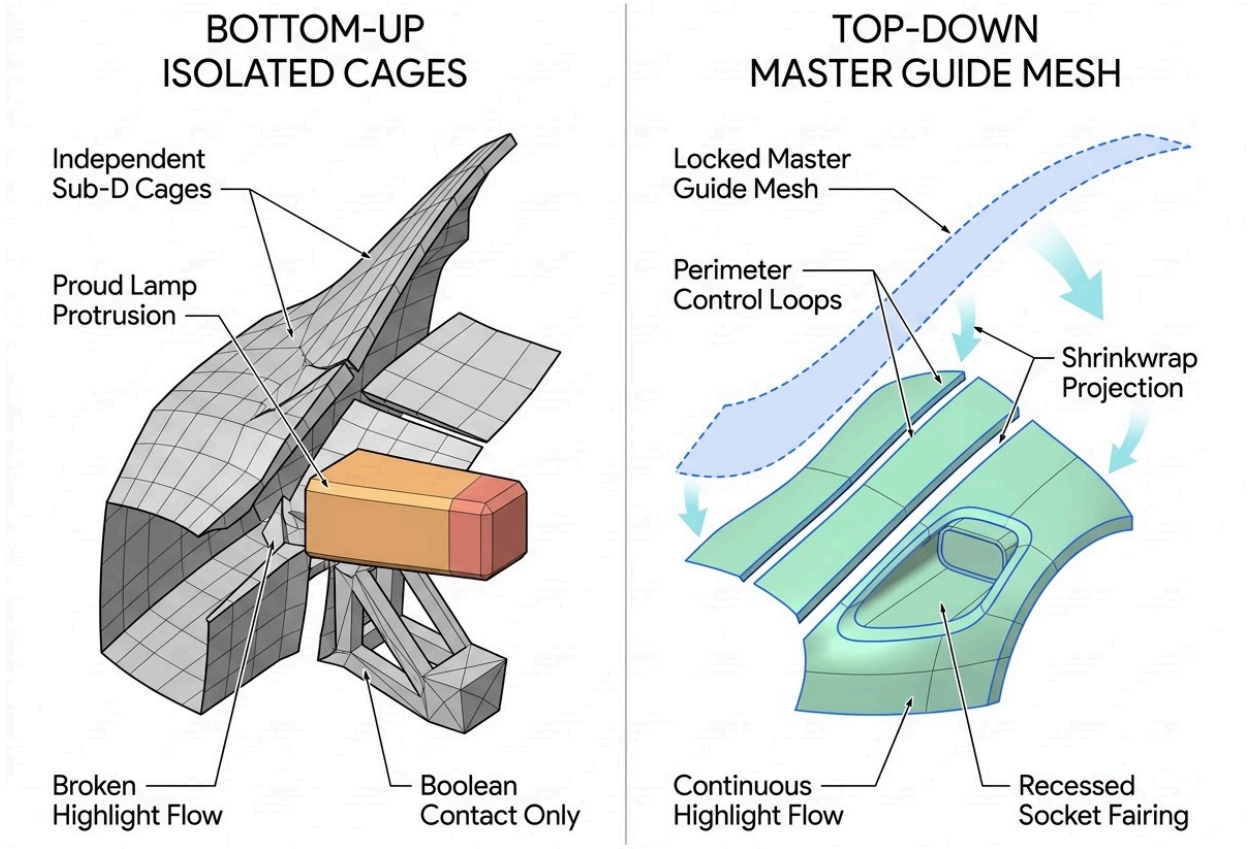
Empirical performance of LLM/VLM 3D pipelines across three diagnostic dimensions. Top: In long-horizon part editing, **self-rollout** causes rapid volumetric drift by Turn 5 compared to ground-truth state anchoring. Middle: Off-the-shelf VLMs perform near random-guess baselines on 3D depth alignment until guided by projected wireframe overlays (**RefineAny3D**). Bottom: Expanding visual inspection to **21 calibrated views** with typed engineering feedback substantially increases geometric IoU and requirement satisfaction.

Data sources: EditHero Benchmark (arXiv:2610.02298), RefineAny3D (arXiv:2608.09147), Hephaestus-CCX (ResearchGate)

As demonstrated in the *EditHero* benchmark evaluating long-horizon part-level 3D editing across models including Opus 5.5 and Sol 6, when an agent iteratively modifies parts under self-rollout (where turn k builds upon the agent's own imperfect state S_{k-1} rather than a locked ground-truth reference state), errors compound rapidly[3]. In 5-turn editing chains, volumetric Intersection-over-Union (IoU) to the target drops from 0.68 to 0.52 for Nano3D and from 0.46 to 0.36 for 3DEditFormer when ground-truth spatial anchoring is removed[3]. Consistent failure analyses across more than ten state-of-the-art LLMs confirm that procedural faithfulness collapses in order-sensitive compositional settings when earlier geometric states are mutated or re-interpreted without hard spatial constraints[4].
Furthermore, the agent could not use its visual review pass (geometryforge review) to correct this drift because the reference photography was uncalibrated to the studio review cameras[1]. When a Vision-Language Model (VLM) inspects an uncalibrated 2D photograph alongside a 3D render taken from a different focal length and camera pose, it cannot accurately judge metric proportions, roof roll, or depth offsets[17]. In the *Refine3D* benchmark, off-the-shelf VLMs such as Qwen3-VL-8B achieve only 28.7% directional accuracy and 11.4% full accuracy on 3D depth and alignment refinement—performing near or below a random-guess baseline ($\sim 28\%$ and $\sim 11\%$) due to systematic monocular depth bias[17]. Moreover, empirical evaluations of VLM visual judges demonstrate that when an execution trace reports successful tool calls (such as 61 passing interface checks), open-weight VLM judges accept 78% to 90% of visibly defective outputs, up from 7% to 19% when tool-success text is withheld[18]. Because GeometryForge fed the agent passing test logs alongside uncalibrated studio renders, the agent's critic loop suffered from confirmation bias and failed to halt on the chunky front lamps or faceted engine bay[1].

Architectural Solution: The Top-Down "Master Guide Mesh" Surfacing Paradigm

To resolve the conflict between monolithic simplicity and part-level fragmentation, the agent's modeling paradigm must be restructured around the industry-standard automotive **Master Model (Class-A Concept Surfacing)** workflow[5], implemented programmatically in Blender via the **Guide Mesh (Shrinkwrap-Below-Subsurf)** technique[6] and in Houdini via **Shared Master SDF/NURBS Skin Projection**[7]. In professional hard-surface vehicle production, artists never model individual body panels, headlamp housings, or skirt fairings as independent free-floating subdivision cages, nor do they attempt to route all mechanical details into a single monolithic mesh[6]. Instead, they decouple **surface curvature authority** from **panel topology and mechanical detailing** across a strict three-tier hierarchy[6].



Tier 1: The Locked Master Guide Mesh (_master_hull)

Before any individual collection (front_body, cabin, skirt, roof) or mechanical detail is authored, the agent must construct a single, continuous, watertight Master Guide Mesh (designated _master_hull in a non-rendering reference collection)[6]. Because _master_hull contains zero cutouts for headlamps, zero panel gaps, and zero mechanical sockets, its control cage requires very few polygons, allowing the agent to maintain clean, evenly spaced quad loops that run uninterrupted from the front bumper across the nose, up the windshield pillars, across the roof roll, and down into the lower skirt[6]. In accordance with established Guide Mesh principles, _master_hull is evaluated at a Subdivision Surface level two levels higher than the final visible panels—specifically Render Level 4 or 5 for _master_hull, compared to Level 2 or 3 for the final child panels[6]. Where sharp styling creases are required, such as the chine line separating the upper hull from the lower skirt, subdivision creasing is applied directly on _master_hull rather than inserting congested support loops[6]. Once _master_hull passes camera-calibrated silhouette and proportion verification against the primary reference image, it is frozen as an immutable Safe-Point Datum Surface that no subsequent --parts run is permitted to alter[1].

Tier 2: Top-Down Panel Decomposition and Shrinkwrap Binding

Rather than lofting front_body, cabin, skirt, and roof from scratch in separate Python functions, the agent must derive every exterior body panel directly from copies or topological slices of

_master_hull[6]. The agent duplicates the relevant face regions of the low-poly _master_hull cage to create individual panel objects (front_fascia, hood_panel, headlamp_bezel, cabin_shell, skirt_panel) and routes the topological cutouts for lamps, vents, and windows directly within these derived cages[1]. To prevent the Subdivision modifier from pulling panel corners inward or distorting cutout boundaries during smoothing, the agent programmatically selects all boundary and cutout edges of each panel and performs a $0.004\ \mathrm{m}$ inset operation (deleting any redundant centerline control loop generated on mirrored panels)[6]. This establishes uniform perimeter control loops around every panel border and lamp opening, after which every exterior body and fairing object is assigned a standardized, non-destructive four-stage modifier stack via bpy.data.objects[panel].modifiers[6].

Modifier Stack Order	Blender Modifier Type	Key Programmatic Parameters (bpy)	Geometric & Aesthetic Function
1 (Top)	SUBSURF	levels = 2, render_levels = 2 (or 3 for closeups)[6]	Smooths the panel's quad topology and cutout control loops *prior* to surface projection[6].
2	SHRINKWRAP	target = _master_hull, wrap_method = 'NEAREST_SURFACEPOINT' (or 'PROJECT'), vertex_group = 'guide_bound' [cite: 6, 25]	Locks all exterior panel vertices onto the Level-4 _master_hull, guaranteeing unbroken G_2 highlight flow across panel cuts and lamp openings[6].
3	SOLIDIFY	thickness = 0.002 to 0.004 ($2–4\ \mathrm{mm}$), offset = -1.0 (inward)[6]	Generates realistic sheet-metal/composite panel thickness inward without puffing out the exterior silhouette[6].
4 (Bottom)	BEVEL	limit_method = 'WEIGHT', width_type =	Applies crisp, uniform micro-fillets along rim and

		'PERCENT', width_pct = 100, segments = 2 [cite: 6]	cutout edges flagged with bevel_weight = 1.0, catching realistic specular edge highlights[6].

The critical mechanism within this stack that resolves both the proud front lamps and the faceted engine socket housing is Shrinkwrap Vertex-Group Modulation[6]. When building a recessed headlamp housing or the painted engine receiving bay (mounts), the outer perimeter loop of the housing is assigned a weight of 1.0 in the guide_bound vertex group, locking the rim 100% flush to _master_hull[6]. As the geometry extrudes inward into the lamp cavity or bearing chamber, the vertex group weight transitions smoothly from 1.0 to 0.0 across the inward control loops[6]. Consequently, the transition from the exterior body into the recessed socket bay becomes a seamless, curvature-continuous (G_2) painted fairing return, inside which the crisp mechanical internals—lenses, emitters, bearing races, support webs, and hose glands—sit recessed rather than protruding proudly outside the vehicle envelope[1].

Tier 3: 3D Interface Rings and Reusable Parametric Toolkit Helpers

To eliminate assembly failures when connecting non-body mechanical parts—such as mating the paired engines to mounts or pads to chassis—GeometryForge must replace 1D scalar plane coordinates ($X = -3884\ \text{mm}$) with explicit 3D Interface Rings (InterfaceRing data-blocks) and verified helper functions[1]. Recent research on agentic CAD reconstruction (*CADForge*) demonstrates that LLMs repeatedly fail at raw compound compositions, fragile filleting, invalid sweeps, and unstable Boolean cuts unless error-prone operations are encapsulated into verified, reusable toolkit functions such as safe_union, strict_cut, and sweep_tube_along_path[8]. Extending this principle to GeometryForge, whenever a parent part (such as the recessed fairing pocket in mounts) defines an attachment opening, it must export an immutable 3D boundary loop object (_interface_engine_mount_L) containing exact vertex positions, inward tangents, and surface normals along the seam[7]. The child sub-assembly (engines or internal bearing race) is prohibited from guessing its placement via parameters.json scalar coordinates; instead, it must invoke a verified helper function that bridges directly from the parent's boundary loop[1]. This guarantees watertight topological alignment and coaxial orientation regardless of how many local iterations the child component undergoes[2].

Upgrading the Visual Critique Loop and Automated Verification Gates

Even with a top-down surfacing architecture, an LLM coding agent will drift if its feedback loop cannot perceive curvature errors, depth misalignments, or proportional discrepancies[2]. Reforming GeometryForge requires replacing qualitative, uncalibrated VLM glances and naive Boolean contact checks with deterministic, multi-modal geometric and visual gates[2].

Camera Pose Calibration and Projected Wireframe Overlays

Before any geometry code is written in a reference-driven study, the Reference Intake stage in SKILL.md must solve and lock the reference camera intrinsics $\mathbf{K}_v$ (focal length, principal point) and extrinsics $\mathbf{T}_v$ (camera rotation and translation) relative to the vehicle's canonical datum axes[1]. Using perspective calibration matching 3D bounding-box corners of the vehicle wheelbase and pad centers to 2D image coordinates, GeometryForge must instantiate a locked Camera_Ref_Primary inside the Blender scene[1]. Rather than asking the VLM critic to judge raw clay renders against an uncalibrated photo, the review pipeline must project 3D wireframe bounding boxes and major section curves (roofline centerline, A-pillar edge, beltline crease, and pad arches) directly onto both the reference image and the candidate render[17]. As proven in *RefineAny3D*, providing a VLM with projected wireframe alignment overlays transforms monocular depth and proportion estimation into a 2D semantic alignment task, boosting directional accuracy from 28.7% to 89.2% and full alignment accuracy from 11.4% to 76.7% [17].

Quantitative Multi-View Visual Metrics and Zebra Striping

In *Hephaestus*, expanding agentic visual inspection from standard 4–6 studio views to 21 calibrated views—comprising orthographic exteriors, perspective camera matches, localized interface close-ups, and internal cross-sectional X-ray cuts—coupled with quantitative engineering feedback raised volumetric IoU from 0.444 to 0.592 and increased requirement pass rates from 38.8% to 60.5% [2]. Similarly, iterative visual alignment frameworks enforce an explicit incremental improvement gate:

$$\Delta \mathrm{IoU}_t = \mathrm{IoU}(M_t, M_{\text{target}}) - \mathrm{IoU}(M_{t-1}, M_{\text{target}}) \geq \tau$$

requiring each revision step to achieve a positive silhouette overlap gain or reach an absolute acceptance threshold ($\mathrm{IoU} >$) before advancing[26].

To cure the agent's blindness to broken highlight flow and surface lumps across front_body, cabin, and skirt, geometryforge review must incorporate a technical Zebra Striping (Isophote) render pass[5]. Because standard Blender viewport Zebra matcaps suffer from pixel stepping at shallow angles and cannot be easily customized in headless batch renders, the review fixture should inject a procedural Cycles/EEVEE Wave-Texture Reflection Shader (Texture Coordinate [Reflection] -> Mapping -> Wave Texture -> Emission) onto all exterior hull collections[29]. In this diagnostic pass, G_0 positional collisions (such as the proud front lamps) cause zebra bands to break and jump discontinuously across the seam; G_1 tangent joints without curvature matching cause sharp kinks in the stripes; and true G_2 Class-A surfaces (achieved via the _master_hull Shrinkwrap workflow) exhibit smooth, unbroken, flowing bands across panel gaps[5].

Region-Masked Edit Preservation and Differential Continuity Checks

When the agent executes a localized run (geometryforge run --parts mounts or --parts front_body), GeometryForge must enforce Non-Edited Region Invariance[3]. Following the formalization in *EditHero*, for a camera view v, let $\Pi_v(S, \mathcal{M}_k)$ denote the depth-tested pixel projection of the target edited part set $\mathcal{M}_k$, excluding pixels occluded by other geometry[3]. With a one-pixel dilation operator $\mathcal{D}$ over the image domain Ω, the active edit region E_k^v and the unedited preservation region U_k^v between state S_{k-1} and state S_k are defined as:

$$E_k^v = \mathcal{D}\left(\Pi_v(S_{k-1}, \mathcal{M}_k) \cup \Pi_v(S_k, \mathcal{M}_k)\right), \qquad U_k^v = \Omega \setminus E_k^v$$

During any localized part iteration, the verification harness must assert that geometric and normal-map variance within the unedited complement U_k^v remains zero outside designated interface blend margins, while evaluating perceptual and silhouette improvements exclusively within E_k^v [3]. Simultaneously, to defeat Goodhart's Law in checks/native-interfaces-final.json, the verification suite must replace generic Boolean contact checks with four typed verification classes that distinguish internal mechanical engagement from exterior Class-A surface continuity[1].

Interface Class	Current GeometryForge Check (Flawed Proxy)	Upgraded Mathematical & Programmatic Verification Gate	Target Acceptance Threshold
Class 1: Exterior Panel & Fairing Seams (front_body $\leftrightarrow$ skin, cabin $\leftrightarrow$ roof, lamp_bezel $\leftrightarrow$ front_body)	Boolean mesh intersection ($\mathrm{Vol}(A \cap B) >$)[1]	**1. Master Surface Deviation**: Max vertex distance of panel exterior faces to _master_hull via BVHTree raycast. **2. Seam Step/Flushness Error**: Normal-projected height offset Δh between opposing boundary loops.	**1.** $<$ **2.** $\|\Delta h\| <$; gap width $2.0–4.0\ \mathrm{mm}$ **3.** $\theta_{\text{normal}} <$ ($\cos\theta >$) **4.** $<$ [cite: 5, 16]

		3. Tangent Continuity (G_1): Normal vector alignment $1 - (\mathbf{n}_A \cdot$ across the panel gap. **4. Curvature Continuity (G_2)**: Relative radius deviation $\|R_1 - R_2\|/(R_1 +$ across boundary vertices[5].	
Class 2: Recessed Housing Transitions (mounts socket bay $\leftrightarrow$ front_body/cabin, front_lamps $\leftrightarrow$ front_body)	Clearance Boolean cut + flange contact check[1]	**1. Rim Flushness**: Perimeter vertex group (weight = 1.0) must have $0.0\ \text{mm}$ step offset to _master_hull. **2. Inward Draft & Fillet Check**: Cavity wall normals must sweep inward continuously ($>$ draft) without exterior protrusion beyond _master_hull envelope[6].	**1.** $0.0\ \text{mm}$ exterior protrusion (zero proud vertices outside _master_hull) **2.** Minimum fillet radius $>$ on painted returns
Class 3: Internal Mechanical Mating (engines collar $\leftrightarrow$ mounts bearing	Mesh intersection check[1]	**1. Coaxial Ring Alignment**: Centroid and principal axis alignment between	**1.** Axis misalignment $<$; centroid offset $<$

race, hoses $\leftrightarrow$ glands, webs $\leftrightarrow$ beams)		parent _interface_ring and child mating boss. **2. Controlled Engagement Lap**: Volumetric penetration depth bounded within [min_lap, max_lap] without piercing exterior Class-A skins[1].	**2.** Lap $= 48.0 \pm$; 0 intersections with exterior skin[1]
Class 4: Global Silhouette & Proportion (Full Vehicle Assembly vs. Calibrated Reference)	Uncalibrated side-by-side visual review[1]	**1. Calibrated 2D Silhouette IoU**: Binary mask IoU against primary reference image through Camera_Ref_Primary. **2. Cross-Sectional Slice Audit**: 21-view render suite including X, Y, Z planar section cuts to verify watertight belly-to-bumper closure[2].	**1.** Silhouette $\text{IoU} >$ [cite: 26] **2.** 0 unsealed boundary edges on lower hull/keel[1]

Comprehensive Action Plan: Re-Engineering the Skill and Execution Pipeline

To guide Opus 5.5 and GPT 6.1 Sol to produce a high-detail, visually coherent hover taxi and future reference-driven hard-surface assets, the project workflow must be refactored from an abrupt two-stage jump (pilot.py to advanced.py) into a Four-Gate Hierarchical State Machine enforced directly by SKILL.md and geometryforge verify[1].

Phase 1: Camera Calibration and 3D Spatial Blueprinting (Gate 0)

Before any polygon mesh is constructed, the agent must establish a unified 3D coordinate contract that extends beyond scalar text entries in parameters.json[2]. The workflow begins by

calibrating Camera_Ref_Primary to the primary reference photograph using known vehicle datums—specifically the four accepted hover pad centers and the overall length and width envelope[1]. Once the camera pose is locked, the agent emits lightweight 3D curve cages representing the primary feature lines (roofline profile, beltline sweep, fender crown, nose-to-skirt chine, and engine mount centerlines) and projects them onto the reference image[17]. Because generated supplementary views (such as rear three-quarter or overhead roof views) frequently contain perspective and proportional inconsistencies[1], the agent is prohibited from modeling directly from a supplementary view until its proportions are normalized to the locked 3D bounding proxy established by the primary reference[5].

Phase 2: Master Guide Mesh Authoring and Curvature Lock (Gate 1)

In the first modeling stage (models/master_surface.py), the agent is prohibited from creating individual parts, lamps, or mechanical details[6]. Instead, it constructs _master_hull, capturing the complete exterior skin of the hover taxi—encompassing the front nose, cabin greenhouse, roof crown, rear-quarter taper, and lower skirt—as a minimal, clean Catmull-Clark control cage evaluated at Subdivision Level 4[6]. To pass Gate 1 verification, GeometryForge renders the binary silhouette mask from Camera_Ref_Primary and computes the 2D Silhouette IoU against the segmented reference vehicle, requiring $\mathrm{IoU} >$ and verifying that the roof roll, pillar widths, cabin crown, and rear-quarter taper align with the projected wireframe guides[1]. Simultaneously, the pipeline renders the Procedural Zebra Stripe Pass across 21 calibrated views to verify smooth, unbroken G_2 highlight lines from nose to skirt and across the cabin transition[2]. Once validated, _master_hull is locked (hide_render = True) and its cryptographic hash is recorded in decisions.json[1].

Phase 3: Top-Down Panel Decomposition and Recessed Bay Pockets (Gate 2)

In the second modeling stage (models/panels_and_bays.py), the agent decomposes _master_hull into the primary structural collections (front_body, cabin, skirt, roof, and the outer fairing of mounts) via topological face-set splitting rather than independent lofting[1]. The agent routes openings for the front lamps, cabin glazing, roof beacons, and engine socket receiving bays, applying the $0.004\ \mathrm{m}$ perimeter inset helper to establish clean quad control loops around every opening and seam before binding the Subsurf -> Shrinkwrap (_master_hull) -> Solidify -> Weighted Bevel stack[1].

Recessed cavities are formed through vertex-group falloff[6]. To fix the proud, chunky front lamps, the agent extrudes the lamp housing inward from the front_body cutout border, setting guide_bound vertex weights to 1.0 on the outer bezel rim and 0.0 on the recessed lens seat[1]. This recessed pocket preserves the sleek nose-to-skirt aerodynamic contour while creating an integrated housing for the internal lamp optics[1]. Similarly, to fix the exposed, faceted engine socket housing in mounts, the agent constructs a smooth, painted Recessed Receiving Fairing derived directly from _master_hull[1]. The outer lip of the socket fairing remains G_2 -locked to the body and roof via the Shrinkwrap modifier, sweeping smoothly inward via a filleted return to a

recessed internal bulkhead ring (_interface_engine_socket)[1]. The structural support webs, crossbeam, and retaining races are positioned behind this painted fairing return so that only the purposeful mechanical mating interface is visible in the bay opening[1]. Finally, the lower boundary loops of front_body and skirt are bridged directly to the longitudinal chassis keel rails using a shared lower datum loop, sealing the open belly-to-bumper transition[1]. Gate 2 verification executes Class 1 and Class 2 surface continuity checks alongside X-ray section cuts to confirm zero proud protrusions and watertight underbody closure[2].

Phase 4: High-Frequency Mechanical Kitbashing and Interface Assembly (Gate 3)

In the final modeling stage (models/mechanical_details.py), with the global silhouette, panel gaps, and recessed bays permanently locked to _master_hull, the agent can safely leverage its strength in modular procedural detailing without risking global form drift[1]. The agent instantiates the preserved four hover pads (181,072 triangles) onto the chassis saddle interfaces, and attaches the explicit quad-lofted engines (61,112 triangles, $7\ \mathrm{mm}$ corner radii, $X = -3884\ \mathrm{mm}$ aft armor lap) and hose glands directly to the _interface_engine_socket data-blocks inside the recessed mount fairings[1]. The recessed front lamp pockets are populated with multi-element reflector optics, internal bezels, and flush crowned outer lenses derived from _master_hull, mirroring the successful technique used on the cabin glazing[1]. Secondary micro-details—including fasteners, emitters, roof sign louvers, and service plates—are projected onto the evaluated parent panels via SHRINKWRAP modifiers or bmesh BVHTree normal projection so that no detail floats or intersects arbitrarily[1]. Gate 3 verification then executes the complete 21-view inspection suite, verifies region-masked invariance (U_{L}^{v}) on all adjacent body panels, and confirms that all Class 1 through Class 4 interface checks pass[2].

Concrete Specification Updates for SKILL.md and the GeometryForge Toolkit

To operationalize this architecture, the following normative rules and helper specifications should be integrated directly into SKILL.md, references/surface-modeling.md, and the GeometryForge Python library[7]:

Target Module	Current Skill / Pipeline Gap	Required Specification Update in GeometryForge
SKILL.md — Reference Intake	Allows modeling to proceed with uncalibrated cameras and inconsistent generated supplementary views[1].	**Mandatory Camera & Proxy Gate**: Before authoring geometry code, solve or approximate a calibrated Camera_Ref_Primary

		matching the primary reference image. Construct a 3D wireframe bounding proxy and overlay its projection onto the reference image (RefineAny3D alignment check)[17]. Subordinate all supplementary views to the primary 3D proxy dimensions; reject or re-prompt supplementary views that violate primary proportions[7].
SKILL.md — Surface Modeling (references/surface-modeling.md)	Advises establishing *"proportions and broad curvature, then panel structure, then mechanical detail"* conceptually, but lacks an enforceable Blender/Houdini topological mechanism, allowing parts to be built as independent cages[1].	**Mandatory Master Guide Mesh Protocol**: For any multi-panel curved vehicle or hard-surface hull, author and freeze a single watertight _master_hull object (Subdivision Level 4) before splitting collections[6]. All exterior body panels (front_body, cabin, skirt, roof, outer mounts fairings) **must** be sliced from _master_hull, inset by $0.004\ \mathrm{m}$ at borders/cutouts for control loops, and bound via a Subsurf -> Shrinkwrap (_master_hull) -> Solidify -> Bevel stack[6]. Independent free-hand lofting of adjacent body panels is strictly prohibited.
SKILL.md — Mechanical Housings & Cutouts	States *"A clearance cut alone is not a mechanical housing"*, which caused the	**Recessed Fairing & Vertex-Group Falloff Rule**: Mechanical sockets, lamp

	agent to build an exposed, faceted mechanical rig (mounts) and proud lamp blocks[1].	clusters, and vents embedded in curved bodywork must never protrude proudly outside _master_hull (unless explicitly seen in reference) or leave raw faceted framing exposed[1]. Form the receiving bay by extruding the panel cutout *inward* while tapering the Shrinkwrap vertex group weight from 1.0 at the flush outer rim to 0.0 at the internal mounting ring[6]. Place internal mechanical structures (races, webs, lenses, glands) inside the recessed cavity behind the painted fairing return[1].
SKILL.md — Revise and Verify (--parts execution)	Encourages running isolated --parts updates with boolean contact checks, leading to multi-turn self-rollout drift and Goodharted intersections[1].	**Interface Ring Binding & Region Invariance**: Prohibit positioning child sub-assemblies via raw scalar coordinates when mating to curved surfaces; require binding to exported 3D _interface_ring data-blocks[8]. Enforce region-masked preservation (U_{κ}^{v}) during --parts runs[3]. Upgrade geometryforge verify to enforce four-tier validation: (1) Silhouette $\mathrm{IoU} >$ on calibrated camera, (2) BVHTree $0.0\ \mathrm{mm}$ proud-protrusion and $<$ normal-tangent checks

		across exterior seams, (3) 21-view render inspection including X-ray section cuts, and (4) Procedural Wave-Texture Zebra stripe continuity[2].
Python Helper Library (geometryforge.helpers)	Supplies basic primitives, leaving complex CAD/Sub-D operations to fragile raw bpy/bmesh synthesis by the LLM[7].	**Verified High-Level Surfacing Helpers**: Provide tested Python functions: build_guide_mesh_stack(panel_obj, guide_obj, vgroup, thickness, bevel_width), inset_panel_perimeter(bm, inset_dist=0.004), extract_interface_ring(obj, edge_loop_tag), recessed_pocket_extrude(obj, loop_id, depth, draft_angle), and render_zebra_diagnostic(scene, camera_list)[6].

By separating macro-curvature authority (_master_hull) from topological panel decomposition (Shrinkwrap-bound panels and inward-drafted fairing pockets) and micro-mechanical detailing (interface-ring-mounted kitbash assemblies), the coding agent is relieved of the impossible task of maintaining global Class-A continuity through raw coordinate algebra across 1,438 separate objects[1]. Simultaneously, coupling camera-calibrated projected wireframe overlays with 21-view Zebra, section-cut, and differential surface-normal checks eliminates Goodharted Boolean intersections and provides the deterministic feedback signal required for autonomous convergence on production-grade 3D geometry[2].

Referenzen

1. README.md
2. Self-Improving CAD Generation Agents with Finite Element Analysis, https://www.researchgate.net/publication/405378679_Self-Improving_CAD_Generation_Agents_with_Finite_Element_Analysis_as_Feedback
3. EditHero: A Benchmark for Long-Horizon Part-Level 3D Editing and, https://arxiv.org/html/2610.02298v1
4. Daily Papers - Hugging Face, https://huggingface.co/papers?q=normal%20refinement
5. Class A surface - Grokipedia, https://grokipedia.com/page/Class_A_surface

6. Essential Guide Mesh Techniques | PDF | Geometry - Scribd, https://www.scribd.com/document/849240770/Full-Guide-to-the-Guide-Mesh
7. SKILL.md
8. CADForge: Agentic Single-View CAD Reconstruction with Explicit, https://www.researchgate.net/publication/415267102_CADForge_Agentic_Single-View_CAD_Reconstruction_with_Explicit_Geometry_Reasoning
9. Can I still save this car's topology? : r/blenderhelp - Reddit, https://www.reddit.com/r/blenderhelp/comments/1u983r4/can_i_still_save_this_cars_topology/
10. Agentic 3D Scene Generation with Spatially Contextualized VLMs, https://arxiv.org/html/2505.20129v1
11. Reward Hacking in the Era of Large Models: Mechanisms, Emergent, https://arxiv.org/html/2604.13602v1
12. Automated Evaluation Systems for LLM and Generative AI Products, https://chierhu.medium.com/automated-evaluation-systems-for-llm-and-generative-ai-products-a-technical-deep-dive-on-metric-7eefa0e119e3
13. AI Safety II: Goodharting and Reward Hacking - Сергей Николенко, https://sergeynikolenko.ru/blog/ai-safety-ii-goodharting-and-reward-hacking
14. How Do Neural Networks Represent the World?, https://publikationen.uni-tuebingen.de/xmlui/bitstream/10900/169628/3/Dissertation_pdfa.pdf
15. Large Language Models: From Internal Architecture and Distributed, https://www.mdpi.com/2076-3417/16/15/7849
16. 5-Step Midjourney to CAD Workflow (With SOP Template), https://innovation-creativity.com/5-step-midjourney-to-cad-workflow-with-sop-template/
17. Depth Refinement as Semantic Alignment for Monocular 3D Detection, https://arxiv.org/html/2608.09147v1
18. Multimodal LLM as an Agent for Unified Image Generation and Editing, https://www.researchgate.net/publication/397201212_GenArtist_Multimodal_LLM_as_an_Agent_for_Unified_Image_Generation_and_Editing
19. Lamborghini Urus. Hi I'm a high-school student and this is ... - Reddit, https://www.reddit.com/r/SolidWorks/comments/vfqbpn/lamborghini_urus_hi_im_a_highschool_student_and/
20. You searched for PTC Creo Parametric - DEVELOP3D, https://develop3d.com/search/PTC+Creo+Parametric/feed/rss2/
21. Computer-aided design-while-engineering technology in top-down, https://www.researchgate.net/publication/282557927_Computer-aided_design-while-engineering_technology_in_top-down_modeling_of_mechanical_product
22. Making a car feels impossible : r/blender - Reddit, https://www.reddit.com/r/blender/comments/1w6jxwb/making_a_car_feels_impossible/
23. 3D Modeling WikiBook | PDF | 3 D Modeling | 3 D Computer Graphics, https://www.scribd.com/doc/256270572/3D-Modeling-WikiBook
24. Place objects on mesh using snap with Python script,

https://blender.stackexchange.com/questions/174727/place-objects-on-mesh-using-snap-with-python-script
25. Object Operators - Blender Python API, https://docs.blender.org/api/current/bpy.ops.object.html
26. Reinforcement Learning in Vision: A Survey - arXiv, https://arxiv.org/html/2508.08189v1
27. GATOR: Generative and Agentic 3D Object Reconstruction ... - arXiv, https://arxiv.org/html/2610.11215v1
28. MedSAM-Agent: Empowering Interactive Medical Image ... - arXiv, https://arxiv.org/html/2602.03320v1
29. blender-render-automation - Agent Skills Library, https://skills.rest/skill/blender-render-automation
30. Using zebra procedural material as replacement for lower quality, https://www.youtube.com/watch?v=hWTsFfOi46Y
31. How to improve MatCap rendering and got rid of stepping at some, https://blender.stackexchange.com/questions/332903/how-to-improve-matcap-rendering-and-got-rid-of-stepping-at-some-angles
32. blender-mcp/BLENDER_SKILLS_REFERENCE.md at main - GitHub, https://github.com/jabbertones-cloud/blender-mcp/blob/main/BLENDER_SKILLS_REFERENCE.md